Cloud-Based Big Data Storage and Processing using Docker Containers

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1. Objective

- To deploy a cloud-based Big Data storage and processing system using Docker containers.
- To enable scalable and portable deployment of Big Data tools.
- To perform distributed data storage and analytics in a containerized environment.

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2. Learning Outcomes

After completing this practical, students will be able to:

- Understand containerization and its role in Big Data deployment.
- Deploy Big Data frameworks using Docker containers.
- Manage distributed storage and processing in cloud environments.
- Perform scalable data analytics using container orchestration concepts.
- Gain practical experience in DevOps and cloud-based Big Data systems.

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3. Software and Hardware Requirements

Software

- Docker Desktop
- Docker Compose
- Cloud platform (AWS / Azure / Google Cloud)
- Big Data tools: Apache Hadoop, Apache Spark
- Linux / Windows OS

Hardware

- Minimum 8GB RAM (recommended 16GB)
- Multi-core processor
- Stable internet connection
- 20GB free storage

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4. Theory

4.1 Big Data in Cloud

Cloud computing enables scalable storage and processing of large datasets without maintaining physical infrastructure.

4.2 Containerization

Containerization packages applications and dependencies into lightweight containers, ensuring consistent deployment across environments.

Docker helps:

- Rapid deployment
- Isolation of services
- Scalability
- Portability

4.3 Docker in Big Data

Docker containers can run distributed Big Data components such as Hadoop NameNode, DataNode, Spark Master, and Worker nodes.

Benefits:

- Simplified cluster setup
- Faster experimentation
- Cost-effective deployment

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5. System Architecture

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              Cloud Infrastructure
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                      ▼
              Docker Container Layer
        ┌──────────────────────────────┐
        │  Hadoop NameNode   Spark Master  │
        │  Hadoop DataNodes  Spark Workers │
        └──────────────────────────────┘
                      |
                      ▼
              Distributed Storage (HDFS)
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                      ▼
               Data Processing (Spark)
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                      ▼
                Analytics & Results
```

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6. Procedure (Step-by-Step)

Step 1: Install Docker

- Install Docker and verify installation.

Step 2: Pull Big Data Docker images

- Pull Hadoop and Spark container images.

Step 3: Configure Docker Compose

- Define services for NameNode, DataNode, Spark Master, and Workers.

Step 4: Start containers

- Use Docker Compose to launch containers.

Step 5: Verify cluster setup

- Access Hadoop UI and Spark UI through browser.

Step 6: Load dataset into HDFS

- Upload sample Big Data files into distributed storage.

Step 7: Run data processing job

- Execute Spark analytics job.

Step 8: Monitor processing

- Observe logs and cluster resource utilization.

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7. Output (Example)

- Hadoop cluster running successfully in Docker containers
- Data stored in distributed HDFS
- Spark job executed with processed output displayed

Example result:

```
Word Count Result:
Data → 1200
Cloud → 850
Docker → 540
```

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8. Result

The Big Data storage and processing system was successfully deployed using Docker containers, enabling scalable and portable cloud-based analytics.

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9. Applications

- Cloud data analytics platforms
- Real-time streaming systems
- Enterprise Big Data pipelines
- Machine learning workflows
- IoT data processing

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10. Advantages

- Easy deployment and scalability
- Reduced infrastructure cost
- High portability
- Faster development and testing
- Efficient resource utilization

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11. Conclusion

This practical demonstrates how Docker containers can be used to deploy cloud-based Big Data systems efficiently. Containerization simplifies cluster setup, improves scalability, and enables portable distributed data processing.

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12. Viva Questions

1. What is containerization and how is it different from virtualization?
2. Why is Docker suitable for Big Data deployment?
3. What is the role of Docker Compose in cluster setup?
4. How does cloud infrastructure improve Big Data scalability?